

Total Marks: 03

Obtained Marks: \_\_\_\_\_

# Discrete Mathematical Structures

## Assignment # 01

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**Instructions:** Copied or shown assignments will be marked zero. Late submissions are not entertained in any case.

(CLO-1)

**Question:**

Let

$p$  : "The student studies regularly."

$q$  : "The student passes the examination."

Write the following statements symbolically:

1. The student studies regularly and passes the examination.
2. The student studies regularly or passes the examination.
3. The student does not study regularly.
4. If the student studies regularly, then the student passes the examination.
5. The student studies regularly if and only if the student passes the examination.

**Solution:**

Given:

$p$ : The student studies regularly.

$q$ : The student passes the examination.

| # | Statement                                                                        | Symbolic Form         |
|---|----------------------------------------------------------------------------------|-----------------------|
| 1 | The student studies regularly and passes the examination.                        | $p \wedge q$          |
| 2 | The student studies regularly or passes the examination.                         | $p \vee q$            |
| 3 | The student does not study regularly.                                            | $\neg p$              |
| 4 | If the student studies regularly, then the student passes the examination.       | $p \rightarrow q$     |
| 5 | The student studies regularly if and only if the student passes the examination. | $p \leftrightarrow q$ |

**Q2:**

**Question:**

Consider:

$$(p \rightarrow q) \wedge (q \rightarrow r)$$

where:

- $p$ : The student studies.
- $q$ : The student understands the concepts.
- $r$ : The student performs well in the examination.

Explain the meaning of the compound proposition in English.

**Solution:**

Given:

**p:** The student studies.

**q:** The student understands the concepts.

**r:** The student performs well in the examination.

In English:

**If the student studies, they understand the concepts, and if they understand the concepts, they perform well in the examination.**